ACTIVE INFERENCE IN THE NEST PROGRAMME · 8 MINUTES

The seam and the audit



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NEST is led by Open Earth Foundation with IIASA, CEU, Imperial College, C40, ICLEI, SCAC and the Active Inference Institute · Schmidt Sciences DEVI

NEST couples energy models from city to nation

Nested Energy System Transitions: a three-year programme from July 2026, funded by Schmidt Sciences through DEVI.

Led by Open Earth Foundation, with IIASA, CEU, Imperial College London, C40, ICLEI, SCAC, the Active Inference Institute.

IIASA builds the two models in this talk: MESSAGEix for the national energy system and STURM for the building stock.

The Active Inference Institute writes the boundaries between the models and the inference layer that crosses them.

The convention is invisible unless declared: one cap, three engines, three carbon prices

ENGINE	WIND, REGION A	CO_2 EMITTED	SHADOW PRICE OF THE CAP
	THE SAME PHYSICS		
OSeMOSYS	4,279 MW	200 kt	$25.5
PyPSA	4,279 MW	200 kt	$28.9
MESSAGEix	4,279 MW	200 kt	$18.6



None is wrong. Each engine annualises capital differently, so each dual is exact for its own objective.

One two-region benchmark, identical inputs, the same binding CO_2 cap (0.20 Mt). Pre-project AII benchmark, 2026.

Each model declares its blanket once, in GNN: what it keeps inside, what it sends, what it receives



STURM's blanket is the boundary and nothing else: prices in, demand out

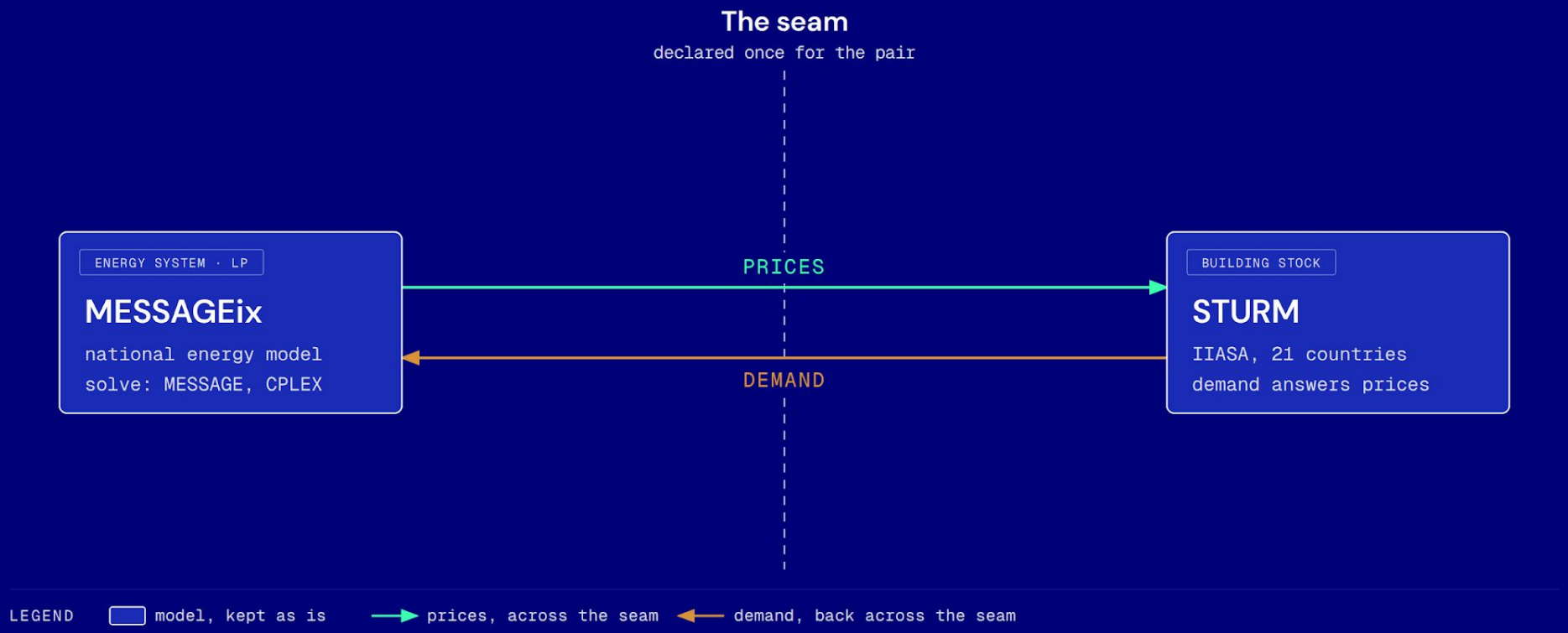


The blanket record is a file anyone can read: the MESSAGEix state space in GNN

AII-D02-messageix-wrapper.gnn.md · NEST-workspace main 26b1cb7 · sha256 410da3416f026830…

```
## StateSpaceBlock
# Sensory — exchanged (inputs crossing the blanket from STURM)
d[2,13,type=float]     # demand per commodity per period
# Sensory — imposed (platform-facing policy lever; never exchanged with STURM)
cap[1,type=float]      # emissions cap
# Active — exchanged (outputs crossing the blanket to STURM)
p[2,13,type=float]     # commodity prices (PRICE_COMMODITY duals)
ef[2,13,type=float]    # emission factors per commodity
# Active — platform-facing (never exchanged with STURM)
e[1,type=float]        # total emissions
lam[1,type=float]      # carbon shadow price (dual of the emissions bound)
J[1,type=float]        # objective value
# Internal (never exchanged; no boundary-file entry, so their units stay here)
K[4,type=float]        # installed capacity per technology [MW]
x[4,13,type=float]     # dispatch per technology per period [MWh]
s[8,1,type=float]      # internal solver state summary (basis, duals machinery)
```

Every variable that crosses the seam carries a unit, a convention, a geography and an owner



Every variable that crosses the seam carries a unit, a convention, a geography and an owner



The record is a file anyone can read: the price entry, with its status words

AII-D02-D03-seam-messageix-sturm.yaml · NEST-workspace main bb3d572 · sha256 b9a168ccd50e4617…

```yaml
pairs:
  prices:
    quantity: energy prices
    sender:   {blanket: AII-D02, variable: p, role: active}
    receiver: {blanket: AII-D03, variable: p, role: sensory}
    direction: messageix -> sturm
    authority: {value: sender, status: declared}
    translation:
      unit:
        value: identity at the boundary; divided by 31.71 inside STURM
        status: observed
        source: aii-run-2026-07-14; message-ix-buildings@6a9cb8b, F10_scenario_runs_MESSAGE_2100.R line 47
      convention:
        value: {carried: "the run's annualisation + discounting, unchanged"}
        status: pending_owner
      temporal_alignment:
        value: run's model years -> STURM 5-year steps, held piecewise-constant
        status: proposed
```

Two feedback loops: MACRO inside MESSAGEix, STURM across the seam



The loop ran scripted and agent-driven, and produced the same files

On 23 October a scripted loop ran the two-way coupling on the Western Europe cutout.

On 28 October, and again on 3 November, an LLM agent ran it from a step-by-step brief: 46 tool calls, every output file identical.

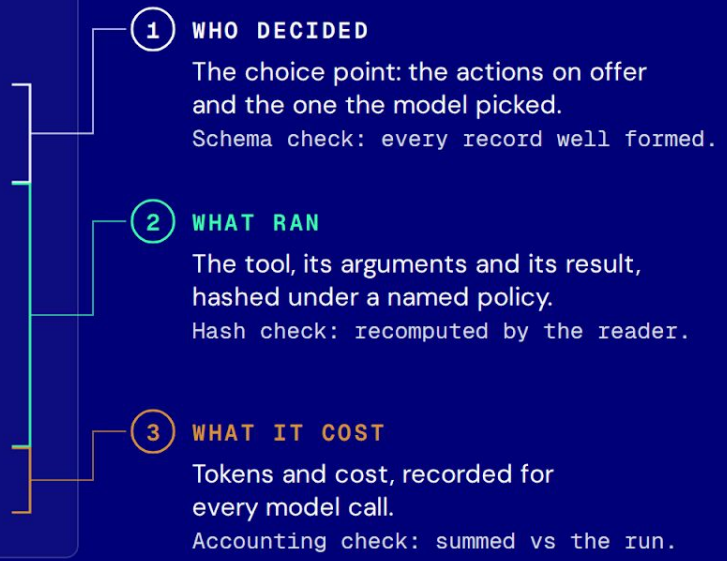
The coupled baseline equals the uncoupled baseline bit for bit. At this seam that is the expected result.

This is a plumbing result. It is not a coupling result, and it is not validation.

Anatomy of one trace record: who decided, what ran, what it cost

EXAMPLE RECORD · agent-trace v0.1 · illustrative, not a run


```
{
  "record_type": "iteration",  "iteration": 7,
  "agent_role": "actor",
  "choice_points": [{
    "candidates": [{"kind": "tool", "descriptor": "solve_message"}, …],
    "selected_indices": [0],  "selection_source": "model" }],
  "tool_calls": [{
    "tool_name": "solve_message",
    "args": {"case": "coupled_baseline_1"},
    "status": "success",  "attempt": 1,
    "result": {"solve_status": "optimal"},
    "hash_policy": "tool-outcome/v1",
    "input_hash":  "0afd6fe9205f7de7…",
    "output_hash": "8c34fd367a5a10ea…" }],
  "token_usage": {"prompt_tokens": 18240, "completion_tokens": 312},
  "cost_usd": 0.0147
}
```

1 **WHO DECIDED**

The choice point: the actions on offer and the one the model picked.

`Schema check: every record well formed.`

2 **WHAT RAN**

The tool, its arguments and its result, hashed under a named policy.

`Hash check: recomputed by the reader.`

3 **WHAT IT COST**

Tokens and cost, recorded for every model call.

`Accounting check: summed vs the run.`

Passing these checks verifies the record. It does not validate the energy computation behind it.

What the first results show, and what they do not

Shown

- Two models written as blankets in one notation, and the seam between them.
- Two real exports pass schema, hash and accounting checks.

Not shown

- That the energy-model computation is right.
- That an agent discovered the coupling on its own.